

Adrian Legaspi

Backend Engineer

Monterrey, Mexico

Profile

Senior Software Engineer with 9+ years delivering production backend services and REST APIs in Node.js, TypeScript, and Python, across enterprise security platforms, fintech, and agentic AI products.

Designs ontologies and semantic data models for enterprise security data, formalized as OWL class and property hierarchies covering domain taxonomies and relationship modeling. Builds LLM applications end to end: RAG retrieval over Chroma vector search, Model Context Protocol integration, agent orchestration, prompt engineering, and fine-tuning workflows, deployed on AWS with Docker and CI/CD. Professional-level English for technical design discussions, documentation, and cross-functional collaboration.

Experience

Software Engineer, JointechAI – Tijuana, Mexico

Aug 2022 – present

- **Viasat Internal Security Application (via Borderux)**: Designed and maintained the semantic data model behind the platform, formalized as an OWL class and property hierarchy: products, the findings raised against them, and the assets that discover those findings. Applied on a government-contract security program; specifics under NDA.
- Delivered Python AWS Lambda functions and REST API services supporting that platform, and built the centralized React and TypeScript dashboard aggregating findings for 4,000+ internal users. Sole frontend engineer embedded in a backend-heavy team of 15.
- **Agentic HR ERP (Jointech AI)**: Tech Lead for a team of 5 engineers on the internal HR platform managing 30 employees and analyzing prospective roles and candidates for hiring. Built its Node.js and Express backend: REST APIs, Model Context Protocol integration, a RAG retrieval pipeline over Chroma vector search, and fine-tuning workflows, plus the React client.

Software Engineer, Arte Digital – Tijuana, Mexico

Jan 2020 – Aug 2022

- **Captira**: Engineered full-stack legal software, building complex Angular data dashboards backed by secure Laravel REST APIs serving 3,000+ legal professionals.
- **Crediverso**: Architected and built full-stack fintech features for a platform serving 10,000 users, using React and a headless WordPress backend. Reduced deployment time by establishing CI/CD pipelines with Codeship.

Web Designer & Developer, LuballSoftware (Freelance) – Tijuana, Mexico

Mar 2016 – Jan 2020

- Developed and managed FireApp, a digital logging application for the Tijuana Fire Department used by 180+ field firefighters and paramedics across 18 stations for pre-hospital incident reports.
- Delivered 25+ web solutions for local businesses over 4 years, ranging from custom e-commerce platforms to landing pages.

Education

Universidad Tecnológica de Tijuana, BS in Tecnologías de la Información – Tijuana, Mexico

May 2015

Skills

Ontologies & Semantic Data: OWL ontologies, semantic data models, taxonomies, class and property hierarchies, domain modeling

Backend & API Engineering: Node.js, Express, TypeScript, Python, REST API design, AWS Lambda, PostgreSQL, Redis, Firebase, Laravel (PHP)

Integration & Async Workflows: Serverless Lambda handlers, Redis, webhook integrations (payment, GitHub), Model Context Protocol

Generative AI & Semantic Search: RAG pipelines, Chroma vector search, embeddings, agent orchestration, prompt engineering, fine-tuning, Gemini APIs

Cloud & DevOps: AWS (S3, EC2, CloudWatch, Lightsail, CodePipeline), Docker, CI/CD (GitHub Actions, CodePipeline), software architecture

Frontend: React, TypeScript, Angular, SCSS

Languages

English: Professional proficiency

Spanish: Native

Side projects

Shroom Merge ([Play Store](#), [App Store](#))

July 2026

- Independently developed and released Shroom Merge (Kinoko Merge on iOS), a mobile merge game, taking the project from concept to Play Store and App Store launch.